FEUP

FACULTY OF ENGINEERING - UNIVERSITY OF PORTO

Systems Engineering

Work Breakdown Structure

SWAID - Plate Resonance (PR)

PLATE RESONANCE
TEAM

Team Members: Afonso Cruz (up202204984)
Alexandre Gorra (up202207016)
Artur Kovbasa (up202208696)
Beatriz Martins (up202108091)
Hugo Gomes (up202208777)
João Oliveira (up202205302)
João Pinto (up202207394)
Lucas Aparício (up202206594)
Miguel Amorim (up202208665)

Proposed by: Professor Sérgio Reis Cunha

Date: April 25, 2026

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1 Introduction

1.1 Document Scope

This document outlines the Preliminary Design for the SWAID Plate Resonance (PR) subsystem, translating its functional architecture into concrete physical and logical solutions. The PR subsystem acts as the translation layer between user interactions and physical actuation, generating synchronous, multi-channel sinusoidal outputs and lighting triggers to drive a 300x200x1mm mirrored stainless steel Chladni plate. To ensure seamless acoustic-visual feedback to the user, the system strictly targets a maximum system-wide latency of 1.0s, with this subsystem (PR) strictly adhering to a processing latency of $\leq$20ms to ensure seamless acoustic-visual feedback.

1.2 Acronyms and Terms

ALSA Advanced Linux Sound Architecture

API Application Programming Interface

COTS Commercial Off-The-Shelf

CP Chladni Plate (and Lighting) Subsystem

DAC Digital-to-Analog Converter

DDS Direct Digital Synthesis

GPIO General-Purpose Input/Output

HI Human Interface Subsystem

JSON JavaScript Object Notation

PR Plate Resonance Subsystem

PWM Pulse Width Modulation

SWAID Standing Wave Acoustic Interactive Demonstrator

TRS Tip-Ring-Sleeve (jacks)

UDS Unix Domain Sockets

ZeroMQ A high-performance asynchronous messaging library

1.3 Document Revision

Table 1: Revision history

Version	Date	Author(s)	Description
1.0	2026-02-27	Lucas Aparício	Template creation
2.0	2026-04-24	Miguel Amorim, João Pinto, Artur Kovbasa, João Oliveira	Final Version

1.4 Structure and Rationale

Our Work Breakdown Structure can be split in different "theaters" of operation such as the ones stated down below:

- **Software Abstraction Layer:** It subdivides in **IPC Communication** and **API Integration**. The first sub-box handles the communication between the HI team and ours, along with inside communications. API integration solves an easy access for our human-machine medium;
- **Chladni Symbol Dictionary:** It subdivides in **Pattern Mapping**, which guarantees the different conditions for the symbols' mapping and **Data Modeling** stablishes a standard for the protocols related to the data flowing through the subsystem;
- **Signal Generation Software: Audio Systems Engine** represents the software coordinating the genesys of audio;
- **Embedded System: Microcontroller** handles the ambient lightning surrounding the plates;
- **Integration & Verification: Component Testing** is responsible to verify the components integrity/performance and **System Validation** check if the system (already merged with the other two subsystems) acts as expected.
- **Project Management: Documentation** pointing to the files needed to fully understand and **Coordination** for both intra-team and inter-team duties and compromises.

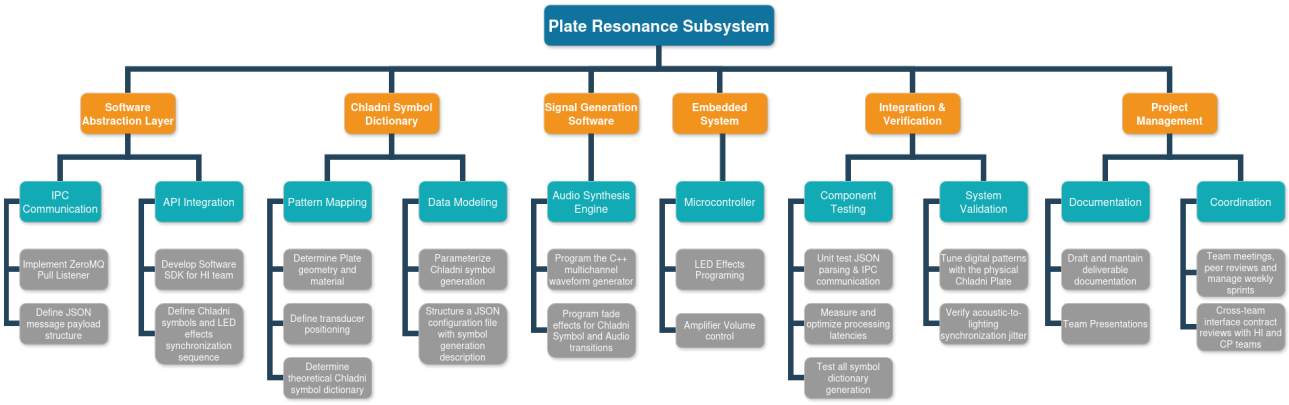

Figure 1

1.5 Work Packages

- **Implement Zero MQ Pull Listener:** Develop a ZeroMQ (ZMQ) receiver node to capture asynchronous, low-latency requests and state changes originating from the Human Interface (HI) subsystem.
- **Define JSON message payload structure:** Establish and formalize a standard JSON schema to serialize and deserialize commands passed between the HI and PR subsystems, detailing parameters like symbol ID, transition duration, and lighting triggers.
- **Develop Software SDK for HI team:** Create a software interface or wrapper that abstracts the underlying network protocols, allowing the HI team to effortlessly send commands to the PR subsystem without dealing with socket complexities.

- **Define Chladni symbols and LED effect synchronization sequence:** Map specific lighting animations and color schemes to their corresponding acoustic resonance states, ensuring a cohesive audiovisual transition during pattern changes.
- **Determine Plate geometry and material:** Collaborate with the Chladni Plate (CP) team to lock down the physical constraints of the plate (e.g., dimensions, shape, metal type) necessary for accurate acoustic wave propagation and node calculations.
- **Define transducer positioning:** Analyze and simulate optimal placement locations for the acoustic transducers underneath the plate to maximize the diversity, symmetry, and clarity of the generated patterns.
- **Determine theoretical Chladni symbol dictionary:** Mathematically and computationally simulate the expected standing wave patterns based on the plate's physical properties, cataloging the theoretical resonant frequencies and modal shapes.
- **Parametrize Chladni symbol generation:** Translate the theoretical modes into actionable signal parameters (frequency, phase, and relative amplitude for each transducer channel) required by the audio engine to reproduce the figures.
- **Structure a JSON configuration file with symbol generation description:** Compile the parametrized signal data into a dynamic, structured JSON configuration file that acts as the "dictionary," allowing the system to load pattern parameters without recompiling the core software.
- **Program the C++ multichannel waveform generator:** Develop the core Audio Systems Engine capable of synthesizing independent, synchronized, and low-latency audio waveforms for multiple transducer channels simultaneously.
- **Program fade effects for Chladni Symbol and Audio transitions:** Implement digital signal processing algorithms to smoothly crossfade frequencies and amplitudes, preventing abrupt mechanical jerks on the plate and ensuring aesthetically pleasing morphs between symbols.
- **LED Effects Programming:** Write the microcontroller firmware responsible for parsing the synchronization commands and driving the ambient lighting around the plate in real-time.
- **Amplifier Volume control:** Develop the software interface and logic to dynamically manage the master gain of the audio amplifiers, protecting the hardware from over-excursion and maintaining safe audible noise levels.
- **Unit test JSON parsing & IPC communication:** Create automated, isolated testing scripts to validate the integrity of the data pipeline, ensuring the system can handle malformed payloads, edge cases, and high-frequency message bursts without crashing.
- **Measure and optimize processing latencies:** Profile the software architecture from message reception to audio output, identifying bottlenecks and optimizing the C++ code to maintain real-time responsiveness.
- **Test all symbol dictionary generation:** Systematically execute every configuration present in the JSON dictionary through the waveform generator to verify that theoretical parameters successfully output the correct electrical signals.
- **Tune digital patterns with the physical Chladni Plate:** Conduct empirical testing with the fully assembled physical plate, tweaking the theoretical frequencies and phases to compensate for real-world material imperfections and structural dampening.

- **Verify accoustic-to-lighting synchronization jitter:** Measure the temporal delay between the acoustic actuation and the visual LED trigger, ensuring the synchronization jitter remains well below the threshold of human perception.
- **Draft and mantain deliverable documentation:** Continuously update the project's formal systems engineering documents (Preliminary Design, baseline architectures, etc.) to reflect the evolving status and final specifications of the PR subsystem.
- **Team Presentation:** Prepare visual materials, structural charts, and speaker notes for mid-term and final project reviews, summarizing subsystem progress, risks mitigated, and project outcomes.
- **Team meetings, peer reviews and manage weekly sprints:** Execute project management duties by planning Agile sprints, distributing workloads fairly, conducting weekly stand-ups, and managing the internal peer-assessment reviews.
- **Cross-team interface contract reviews with HI and CP teams:** Organize and mediate regular alignment meetings with the Human Interface and Chladni Plate teams to validate interface control documents (ICDs), review hardware limits, and ensure seamless system integration.

2 Gantt Diagram

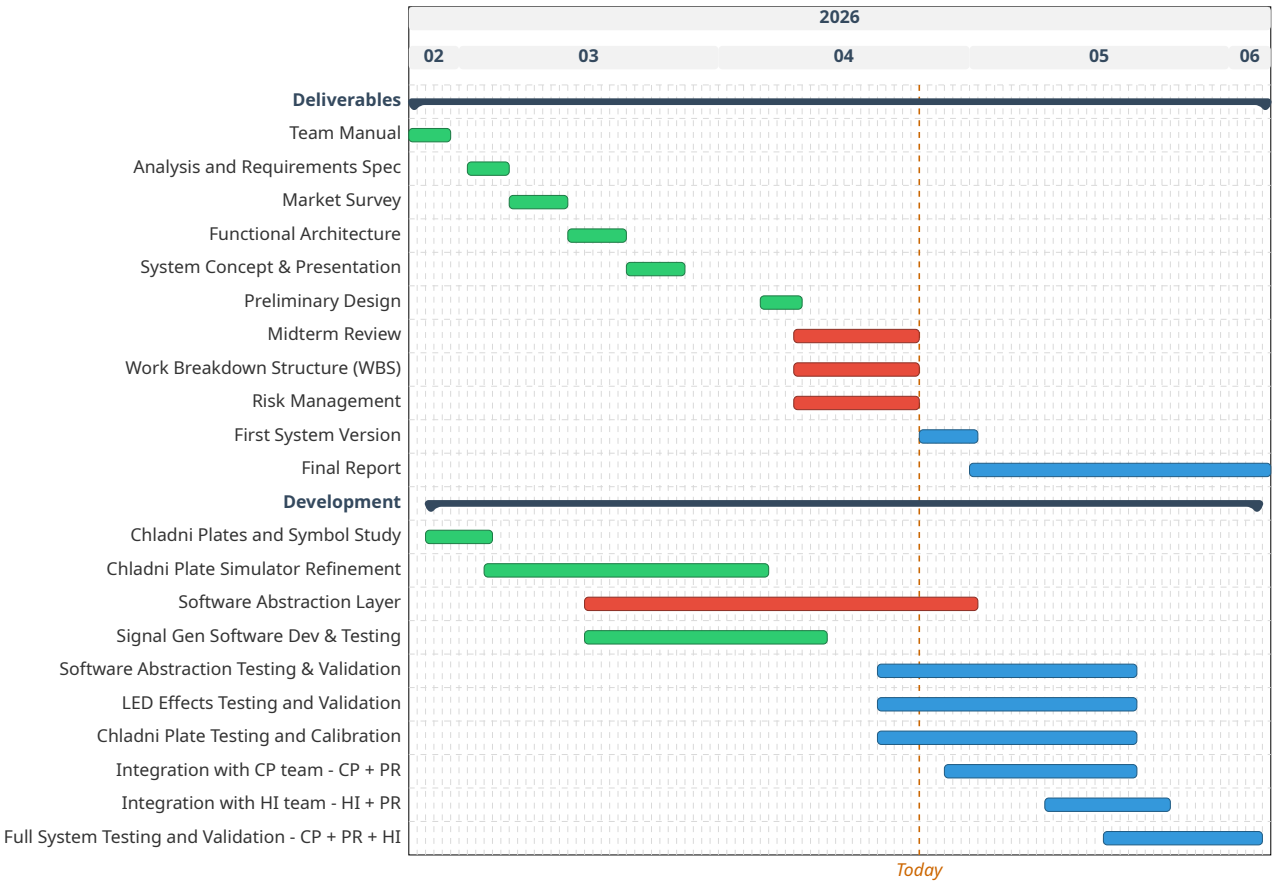

Figure 2: Project schedule overview.

3 Cost Analysis

The SWAID project was assigned a global budget of 480€, distributed equally among the three collaborative teams, resulting in an initial allocation of 160€ per subsystem. Because the hardware requirements for the Plate Resonance (PR) team were inherently low—restricted primarily to a USB sound card and a Raspberry Pi Pico microcontroller—the subsystem operated significantly under budget. To optimize the overall financial efficiency of the project, the PR team's surplus funds were reallocated to the Chladni Plate (CP) team, which faced a heavier financial burden due to costly structural and actuation components.

For the PR team's direct acquisitions, a 7.1 channel sound card with a standard 48kHz sampling rate was determined to be technically sufficient for multi-channel signal generation. Logistical constraints were also heavily weighed; thus, the most affordable in-stock option with a delivery lead time of under two weeks was selected to avoid development bottlenecks. In addition to managing its own budget, the PR team contributed to the project's overall cost-efficiency by defining the specifications for the physical plate. A 300x200mm stainless steel sheet with a 1mm thickness and a factory mirror-polished finish was strategically chosen. Theoretical simulations identified 1mm as the mechanical sweet spot for steel resonance, while bounding the plate to 300x200mm prevented unwarranted material expenses, as larger dimensions offered no substantial improvements in modal pattern reproduction. Furthermore, procuring a pre-polished sheet mitigated the need for secondary surface finishing, thereby reducing hidden labor costs and accelerating the hardware development cycle.

4 Mid-Term Team Evaluation

Table 2: Plate Resonance - Team Evaluation

Name	Peer Evaluation	Absences	Final Grade	Normalized Adjustment
Afonso Cruz	15,0	0	20,0	+0.45
Alexandre Gorra	14,8	0	19,8	+0.17
Artur Kovbasa	15,0	0	20,0	+0.45
Beatriz Martins	14,9	2	18,6	-1.50
Hugo Gomes	14,9	1	19,3	-0.53
João Oliveira	15,0	0	20,0	+0.45
João Pinto	15,0	1	19,4	-0.39
Lucas Aparicio	15,0	0	20,0	+0.45
Miguel Amorim	15,0	0	20,0	+0.45